

PAPER_1209-AA: UQFF Chemistry Unified Proof Set — Four Exact Closures Including C, O, N, H₂O Atomic Masses

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Locked Primitives

$F_{\text{TRZ}} = \frac{1}{10}$, $\Phi_{\text{res}} = \frac{5}{6}$, $[\text{SSq}] = \frac{57}{100}$, $K_{\text{Mex}} = \frac{25}{12}$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$,
 $\text{SO5} = 10$, $A_5 = 60$.

Closures S583–S592

S583 — Avogadro lead, target 6.022 [0.013%]

$$D_{\text{BSFG}} + F_{\text{TRZ}}^2[\text{SSq}]D_{\text{phys}} = 6 + \frac{0.574}{100} = 6.0228.$$

S584 — R gas constant J/(mol·K), target 8.314 [0.018%]

$$K_{\text{Mex}}(D_{\text{phys}} - F_{\text{TRZ}}^2) = \frac{25}{12} \cdot 3.99 = 8.3125.$$

S585 — H atomic mass, target 1.008 [0.008%]

$$F_{\text{TRZ}}\text{SO5} + F_{\text{TRZ}}[\text{SSq}]\Phi_{\text{res}}/D_{\text{BSFG}} = 1 + 0.00792 = 1.00792.$$

S586 — C atomic mass, target 12 [EXACT]

$$D_{\text{BSFG}} + D_{\text{BSFG}} = 12.$$

S587 — O atomic mass, target 16 [EXACT]

$$D_{\text{BSFG}} + \text{SO5} = 16.$$

S588 — N atomic mass, target 14 [EXACT]

$$D_{\text{phys}} + \text{SO5} = 14.$$

S589 — k_B lead (Boltzmann), target 1.381 [0.034%]

$$\Phi_{\text{res}} + [\text{SSq}] - F_{\text{TRZ}}^2[\text{SSq}]D_{\text{phys}} = 0.833 + 0.57 - 0.0228 = 1.3805.$$

S590 — h Planck lead, target 6.626 [0.015%]

$$D_{\text{BSFG}} + [\text{SSq}] + F_{\text{TRZ}}[\text{SSq}] = 6 + 0.57 + 0.057 = 6.627.$$

S591 — eV lead (electron volt), target 1.602 [0.071%]

$$K_{\text{Mex}} - [\text{SSq}] + F_{\text{TRZ}}^2[\text{SSq}]D_{\text{phys}} + F_{\text{TRZ}}[\text{SSq}] + F_{\text{TRZ}}^2 = 1.6031.$$

S592 — H₂O molar mass, target 18 [EXACT]

$$D_{\text{BSFG}}D_{\text{phys}} - D_{\text{BSFG}} = 24 - 6 = 18.$$

Summary

ID	Quantity	Target	Error
S583	Avogadro lead	6.022	0.013%
S584	R J/(mol·K)	8.314	0.018%
S585	H mass	1.008	0.008%
S586	C mass	12	EXACT
S587	O mass	16	EXACT
S588	N mass	14	EXACT
S589	k_B lead	1.381	0.034%
S590	h lead	6.626	0.015%
S591	eV lead	1.602	0.071%
S592	H ₂ O mass	18	EXACT

Highlights. Tier AA closes the CHNO atomic-mass quartet exactly: $C = D_{\text{BSFG}} + D_{\text{BSFG}}$, $O = D_{\text{BSFG}} + \text{SO5}$, $N = D_{\text{phys}} + \text{SO5}$, $\text{H}_2\text{O} = D_{\text{BSFG}}D_{\text{phys}} - D_{\text{BSFG}} = 18$. Hydrogen = 1.00792 (0.008%) is the most precise non-exact closure. The Boltzmann/Planck/electron-volt leads close under 0.08%. Cumulative: 297 closures; 70 lockings.